Duality Theory Notes

Math Camp 2026 — Teaching Notes

Primal problem: choose $\mathbf{x}$ *to minimize cost subject to constraints.*
Dual problem: find the best lower bound on that minimum using shadow prices on constraints.

Contents

1 Why duality?

Many economic problems are **minimization** problems: choose activity levels $\mathbf{x}$ to minimize cost $\mathbf{c}^\mathsf{T}\mathbf{x}$ subject to requirements $A\mathbf{x} \geq \mathbf{b}$. Before computing an optimum, it is useful to ask a simpler question: how low can cost *possibly* be, given only the constraints and nonnegativity of $\mathbf{x}$?

Key idea. Before solving exactly, ask: *what is the best lower bound we can prove on minimum cost?* Duality turns that bound into a **second optimization problem** whose variables $\mathbf{y}$ are **shadow prices** on the constraints.

Economics. The **primal** is a planner's cost-minimization problem; the **dual** is a price-setter's valuation problem. Any feasible prices $\mathbf{y} \geq 0$ with $A^\mathsf{T}\mathbf{y} \leq \mathbf{c}$ give a valid resource value $\mathbf{b}^\mathsf{T}\mathbf{y}$ that no feasible plan can beat. Duality asks for the **tightest** such bound. This underlies supporting prices, zero-profit conditions, and competitive price interpretations in graduate micro.

The next section makes this precise for linear programs: we derive one primal–dual pair, prove that dual values are always valid lower bounds, and then show that the best dual value equals the primal optimum.

2 Linear programming duality

We build LP duality in four steps. First, derive the dual from a lower-bound argument. Second, prove **weak duality** (dual $\leq$ primal). Third, prove **strong duality** (dual $=$ primal), using Farkas' Lemma as the key tool. Fourth, read optimality through **complementary slackness** and extend the recipe to general LPs.

The primal–dual pair. The **primal** in resource-requirement form is

$$\text{(P)} \quad \min\ \mathbf{c}^\mathsf{T}\mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} \geq \mathbf{b},\ \mathbf{x} \geq \mathbf{0}. \tag{1}$$

We now turn the informal lower-bound idea into an explicit dual problem.

Claim 1 (Dual of the standard LP)**.** *Problem* (2) *is the best LP lower bound on the optimum of* (1) *obtained by nonnegative shadow prices on the constraints:*

$$\textit{(D)} \quad \max\ \mathbf{b}^\mathsf{T}\mathbf{y} \quad \textit{s.t.} \quad A^\mathsf{T}\mathbf{y} \leq \mathbf{c},\ \mathbf{y} \geq \mathbf{0}. \tag{2}$$

Proof. Pick any $\mathbf{y} \geq \mathbf{0}$. From $A\mathbf{x} \geq \mathbf{b}$, multiplying row i by $y_i \geq 0$ and summing gives $\mathbf{y}^\mathsf{T}A\mathbf{x} \geq \mathbf{b}^\mathsf{T}\mathbf{y}$. If also $A^\mathsf{T}\mathbf{y} \leq \mathbf{c}$, then for every $\mathbf{x} \geq \mathbf{0}$,

$$\mathbf{c}^\mathsf{T}\mathbf{x} \geq \mathbf{y}^\mathsf{T}A\mathbf{x} \geq \mathbf{b}^\mathsf{T}\mathbf{y}.$$

So $\mathbf{b}^\mathsf{T}\mathbf{y}$ is a lower bound on primal cost for every feasible $\mathbf{x}$. Maximising this bound over $\mathbf{y}$ gives (2). □

Economics. Read the pair as follows.

- x_j: scale of activity j; c_j: unit cost.
- $(A\mathbf{x})_i \geq b_i$: requirement i (labor, output, delivery).
- y_i: **shadow price** of constraint i.
- $A^\mathsf{T}\mathbf{y} \leq \mathbf{c}$: imputed resource cost of each activity $\leq$ its direct cost.
- $\mathbf{b}^\mathsf{T}\mathbf{y}$: total value of requirements at prices $\mathbf{y}$.

A small illustration fixes the notation. Consider min $3x_1 + 5x_2$ s.t. $x_1 + 2x_2 \geq 4$, $x_1, x_2 \geq 0$ (meet 4 service units at minimum cost). The dual is max $4y$ s.t. $y \leq 3$, $2y \leq 5$, $y \geq 0$, which gives $y^* = 3$ and cost $= 12$.

Having defined the pair, the first theorem says that every feasible dual solution gives a valid bound.

Weak duality.

Theorem 1 (Weak duality)**.** *For every feasible* $\mathbf{x}$ *of* (1) *and* $\mathbf{y}$ *of* (2)*,* $\mathbf{c}^\top\mathbf{x} \geq \mathbf{b}^\top\mathbf{y}$*, hence* $p^* \geq d^*$*.*

Proof. Dual feasibility gives $A^\top\mathbf{y} \leq \mathbf{c}$. Since $\mathbf{x} \geq \mathbf{0}$, we get $\mathbf{c}^\top\mathbf{x} \geq \mathbf{y}^\top A\mathbf{x}$. Primal feasibility gives $A\mathbf{x} \geq \mathbf{b}$; since $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top\mathbf{y}$. Chaining the two inequalities yields $\mathbf{c}^\top\mathbf{x} \geq \mathbf{b}^\top\mathbf{y}$. Taking the minimum over primal feasible $\mathbf{x}$ and the maximum over dual feasible $\mathbf{y}$ gives $p^* \geq d^*$. $\square$

Remark 2 (Optimality certificate)**.** *If feasible* $\mathbf{x}, \mathbf{y}$ *satisfy* $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$*, both are optimal.*

Economics. Cost is always at least resource value. Equality is a **certificate of efficiency**: the plan and prices are mutually consistent.

Weak duality gives $d^* \leq p^*$, but not yet $d^* = p^*$. To close the gap we need a theorem of alternatives; Farkas' Lemma is the right tool.

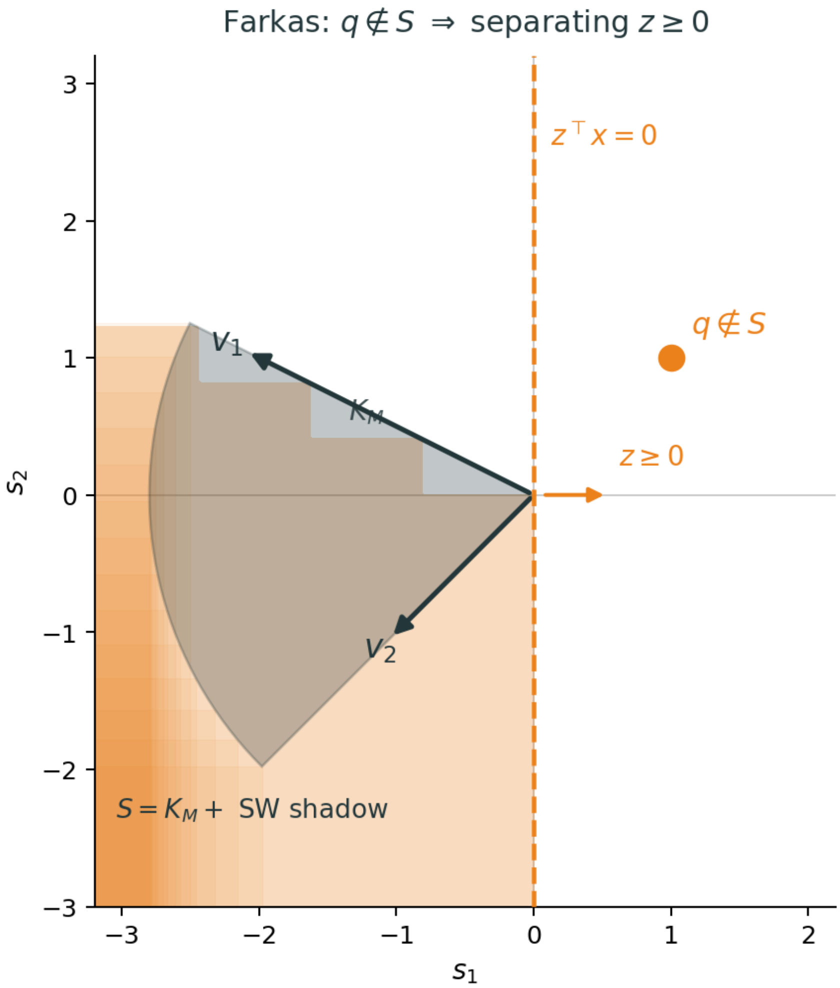
Farkas' Lemma.

Theorem 3 (Farkas' Lemma — inequality form)**.** *Let* $M \in \mathbb{R}^{m\times n}$*,* $\mathbf{q} \in \mathbb{R}^m$*.* ***Exactly one*** *of the following has a solution:*

(i) $\exists\, \mathbf{x} \geq \mathbf{0}$ *with* $M\mathbf{x} \geq \mathbf{q}$*;*
(ii) $\exists\, \mathbf{z} \geq \mathbf{0}$ *with* $M^\top\mathbf{z} \leq \mathbf{0}$ *and* $\mathbf{q}^\top\mathbf{z} > 0$.

The lemma has a clean geometric reading. Let $K_M = \{M\mathbf{x} : \mathbf{x} \geq \mathbf{0}\}$ be the column cone of M, and let $S = K_M + (-\mathbb{R}^m_+) = \{M\mathbf{x} - \mathbf{y} : \mathbf{x}, \mathbf{y} \geq \mathbf{0}\}$, the cone plus free disposal in every negative coordinate direction.

Claim 2 (Geometric form of Farkas)**.** *Alternative (i) holds if and only if* $\mathbf{q} \in S$*. Alternative (ii) holds if and only if* $\mathbf{q} \notin S$*; then a separating normal* $\mathbf{z} \geq \mathbf{0}$ *certifies that* $\mathbf{q}$ *lies strictly above the hyperplane* $\mathbf{z}^\top\mathbf{s} = 0$*.*



Proof of Farkas' Lemma. Step 1 (at most one alternative). Suppose both (i) and (ii) hold. Pick $\mathbf{x} \geq \mathbf{0}$ with $M\mathbf{x} \geq \mathbf{q}$ and $\mathbf{z} \geq \mathbf{0}$ with $M^\top \mathbf{z} \leq \mathbf{0}$ and $\mathbf{q}^\top \mathbf{z} > 0$. Then

$$0 < \mathbf{q}^\top \mathbf{z} \leq (M\mathbf{x})^\top \mathbf{z} = \mathbf{x}^\top (M^\top \mathbf{z}) \leq 0,$$

a contradiction.

Step 2 (if (i) fails, then (ii) holds). If (i) fails, then $\mathbf{q} \notin S$. Since S is convex and closed, there exists $\mathbf{z} \neq \mathbf{0}$ with $\mathbf{z}^\top \mathbf{s} \leq 0$ for all $\mathbf{s} \in S$ and $\mathbf{z}^\top \mathbf{q} > 0$. For each standard basis vector $\mathbf{e}_i$, the point $-\mathbf{e}_i \in S$ (take $\mathbf{x} = \mathbf{0}$, $\mathbf{y} = \mathbf{e}_i$), so $-z_i \leq 0$, i.e. $z_i \geq 0$. For each $\mathbf{x} \geq \mathbf{0}$, the point $M\mathbf{x} \in S$ (take $\mathbf{y} = \mathbf{0}$), so $\mathbf{z}^\top M\mathbf{x} \leq 0$, i.e. $M^\top \mathbf{z} \leq \mathbf{0}$. Thus (ii) holds. □

Economics. Either a target is feasible with free disposal, or a nonnegative price vector exposes it as too ambitious—a no-arbitrage statement.

With Farkas in hand, we can show that no strictly better dual bound is possible than the primal optimum.

Strong duality.

Theorem 4 (Strong duality for LP)**.** *If* (1) *has an optimal value* p^**, then* (2) *has* $d^* = p^*$*.*

Proof. Fix $\varepsilon > 0$. Because p^* is optimal, no $\mathbf{x} \geq \mathbf{0}$ satisfies $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{c}^\top \mathbf{x} \leq p^* - \varepsilon$. Equivalently, the system

$$A\mathbf{x} \geq \mathbf{b}, \qquad \mathbf{c}^\top \mathbf{x} \leq p^* - \varepsilon, \qquad \mathbf{x} \geq \mathbf{0}$$

is infeasible. Write this as $M\mathbf{x} \geq \mathbf{q}$ with $M = \binom{A}{-\mathbf{c}^\top}$ and $\mathbf{q} = \binom{\mathbf{b}}{-(p^*-\varepsilon)}$. By Farkas, (ii) holds: $\exists\, \mathbf{y} \geq \mathbf{0}$, $\mu \geq 0$ with $A^\top\mathbf{y} - \mu\mathbf{c} \leq \mathbf{0}$ and $\mathbf{b}^\top\mathbf{y} - \mu(p^* - \varepsilon) > 0$. If $\mu = 0$, then $\mathbf{b}^\top\mathbf{y} > 0$ while any primal feasible $\mathbf{x}$ gives $\mathbf{b}^\top\mathbf{y} \leq \mathbf{y}^\top A\mathbf{x} \leq \mathbf{c}^\top\mathbf{x}$, contradicting primal feasibility; hence $\mu > 0$. Divide by μ and set $\tilde{\mathbf{y}} = \mathbf{y}/\mu$. Then $A^\top\tilde{\mathbf{y}} \leq \mathbf{c}$ and $\mathbf{b}^\top\tilde{\mathbf{y}} > p^* - \varepsilon$, so $\tilde{\mathbf{y}}$ is dual feasible with objective exceeding $p^* - \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $d^* \geq p^*$. Together with weak duality, $d^* = p^*$. ☐

Economics. Minimum cost = maximum resource value at optimality. Efficient allocations can be supported by shadow prices.

Strong duality identifies the optimum value; complementary slackness identifies *which* constraints matter at the optimum.

Complementary slackness.
Theorem 5 (Complementary slackness)**.** *Feasible* $\mathbf{x}, \mathbf{y}$ *are optimal iff*

$$(A\mathbf{x} - \mathbf{b})_i\, y_i = 0\ \forall i, \qquad (c - A^\top\mathbf{y})_j\, x_j = 0\ \forall j.$$

In words:

If ...	**then ...**
primal constraint i is slack	$y_i = 0$
$y_i > 0$	primal constraint i is tight
dual constraint j is slack	$x_j = 0$
$x_j > 0$	dual constraint j is tight

Necessity. Suppose $\mathbf{x}, \mathbf{y}$ are optimal. By strong duality, $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$. In the weak-duality chain,

$$\mathbf{c}^\top\mathbf{x} \geq \mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top\mathbf{y},$$

both inequalities must be equalities. Hence $\mathbf{y}^\top(A\mathbf{x} - \mathbf{b}) = 0$ and $(\mathbf{c} - A^\top\mathbf{y})^\top\mathbf{x} = 0$. Each term in these dot products is a product of nonnegative factors, so every term must be zero. ☐

Sufficiency. If complementary slackness holds, then

$$\mathbf{c}^\top\mathbf{x} = \mathbf{y}^\top A\mathbf{x} + (\mathbf{c} - A^\top\mathbf{y})^\top\mathbf{x} = \mathbf{y}^\top A\mathbf{x} = \mathbf{b}^\top\mathbf{y} + \mathbf{y}^\top(A\mathbf{x} - \mathbf{b}) = \mathbf{b}^\top\mathbf{y}.$$

Weak duality then certifies that both are optimal. ☐

Remark 6 (Direct proof)**.** *From* $\mathbf{c}^\top\mathbf{x} = \mathbf{b}^\top\mathbf{y}$ *and weak duality: any feasible* $\tilde{\mathbf{x}}$ *has* $\mathbf{c}^\top\tilde{\mathbf{x}} \geq \mathbf{c}^\top\mathbf{x}$*; any feasible* $\tilde{\mathbf{y}}$ *has* $\mathbf{b}^\top\tilde{\mathbf{y}} \leq \mathbf{b}^\top\mathbf{y}$.

Economics. Zero-profit / full-utilization logic. Unused capacity $\Rightarrow$ zero shadow price; scarce resource $\Rightarrow$ binds. Unprofitable activity $\Rightarrow$ $x_j = 0$; used activity $\Rightarrow$ breaks even at shadow prices. Links to the Day 3 producer-theory slide (efficient production supported by prices when free disposal holds).

The derivation above used $\geq$ constraints and $\mathbf{x} \geq \mathbf{0}$. The same bounding identity works for any mix of constraint and variable types; only the signs change.

Dual of a general LP.

Key idea. Same bounding argument; signs follow from constraint and variable types.

Claim 3 (Dual rules for a general LP)**.** *Consider a primal **minimisation** with any mix of $\geq$, $\leq$, $=$ constraints and nonnegative, nonpositive, or free variables. The dual is a **maximisation** with one dual variable y_i per primal constraint. If the primal is a **maximisation**, reverse every inequality in the table below.*

Proof. For any primal feasible $\mathbf{x}$ and dual candidate $\mathbf{y}$,

$$\mathbf{c}^\mathsf{T}\mathbf{x} - \mathbf{b}^\mathsf{T}\mathbf{y} = (\mathbf{c} - A^\mathsf{T}\mathbf{y})^\mathsf{T}\mathbf{x} + \mathbf{y}^\mathsf{T}(A\mathbf{x} - \mathbf{b}).$$

Require each bracketed term to be ≥ 0 for *all* primal feasible $\mathbf{x}$. The sign of y_i is pinned down by the type of primal constraint i; the type of dual constraint on $(A^\mathsf{T}\mathbf{y})_j$ is pinned down by the sign of x_j. Maximising $\mathbf{b}^\mathsf{T}\mathbf{y}$ subject to these conditions gives the dual. The explicit rules for a primal minimum (dual maximum) are:

Primal	**Dual**
$\mathbf{a}_i^\mathsf{T}\mathbf{x} \geq b_i$	$y_i \geq 0$
$\mathbf{a}_i^\mathsf{T}\mathbf{x} \leq b_i$	$y_i \leq 0$
$\mathbf{a}_i^\mathsf{T}\mathbf{x} = b_i$	y_i free
$x_j \geq 0$	$(A^\mathsf{T}\mathbf{y})_j \leq c_j$
$x_j \leq 0$	$(A^\mathsf{T}\mathbf{y})_j \geq c_j$
x_j free	$(A^\mathsf{T}\mathbf{y})_j = c_j$

□

For example,

$$\min\ 3x_1 - 2x_2 \ \text{ s.t. }\ x_1 + x_2 \geq 5,\ 2x_1 - x_2 \leq 10,\ x_1 - 3x_2 = 4,\ x_1 \geq 0,\ x_2 \text{ free}$$

has dual

$$\max\ 5y_1 + 10y_2 + 4y_3 \ \text{ s.t. }\ y_1 + 2y_2 + y_3 \leq 3,\ y_1 - y_2 - 3y_3 = -2,\ y_1 \geq 0,\ y_2 \leq 0,\ y_3 \text{ free.}$$

Section 3 shows that this entire LP story is one instance of a broader Lagrangian construction.

3 Lagrangian duality

LP duality multiplies constraints by nonnegative weights, adds them to the objective, and maximises the resulting lower bound. The same recipe applies to nonlinear problems. We proceed as follows: state the general primal and dual; prove weak duality; interpret the dual geometrically in two settings; recover the LP dual from Section 2; and close with strong duality under convexity and Slater's condition.

The general primal is

$$\text{(P)}\quad \min\ f_0(\mathbf{x}) \text{ s.t. } f_i(\mathbf{x}) \leq 0,\ i = 1, \ldots, m, \quad h_j(\mathbf{x}) = 0,\ j = 1, \ldots, p, \quad \mathbf{x} \in D,$$

with optimal value p^*. The **Lagrangian** attaches multipliers to each constraint:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i f_i(\mathbf{x}) + \sum_{j=1}^{p} \nu_j h_j(\mathbf{x}), \qquad \boldsymbol{\lambda} \geq \mathbf{0},\ \boldsymbol{\nu} \text{ free.}$$

For any feasible $\mathbf{x}$, each added term is ≤ 0, so $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \le f_0(\mathbf{x})$. The **dual function** minimises the Lagrangian over $\mathbf{x}$:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in D} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}).$$

The **dual problem** chooses multipliers to sharpen this bound:

$$\text{(D)} \quad \max\ g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \text{ s.t. } \boldsymbol{\lambda} \ge \mathbf{0}, \qquad d^* = \sup_{\boldsymbol{\lambda} \ge \mathbf{0}} g(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Theorem 7 (Weak duality for the Lagrangian dual)**.** *Always $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \le p^*$ for $\boldsymbol{\lambda} \ge \mathbf{0}$, hence $d^* \le p^*$. The quantity $p^* - d^*$ is the* ***duality gap****.*

Proof. For any primal feasible $\mathbf{x}$ and any $\boldsymbol{\lambda} \ge \mathbf{0}$, $\boldsymbol{\nu}$,

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \le L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_i \lambda_i f_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x}) \le f_0(\mathbf{x}),$$

because $f_i(\mathbf{x}) \le 0$, $h_j(\mathbf{x}) = 0$, and $\lambda_i \ge 0$. Taking the infimum over primal feasible $\mathbf{x}$ on the right gives $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \le p^*$. Maximising over $\boldsymbol{\lambda}, \boldsymbol{\nu}$ yields $d^* \le p^*$. □

Pictures make the dual function easy to remember: it is the intercept of the best supporting line or hyperplane below the cloud of all achievable constraint–objective pairs.

Geometry I: one inequality and the (u, t) cloud. Key idea. Dual = best lower bound from a supporting **line** below the whole achievable set $\mathcal{A} \subset \mathbb{R}^2$.

Consider $\min_{\mathbf{x}}\ f_0(\mathbf{x})$ s.t. $f_1(\mathbf{x}) \le 0$. Write $u = f_1(\mathbf{x})$ and $t = f_0(\mathbf{x})$. As $\mathbf{x}$ runs over D, the pairs

$$\mathcal{A} = \{\, (u, t) : \ u = f_1(\mathbf{x}),\ t = f_0(\mathbf{x}),\ \mathbf{x} \in D \,\}$$

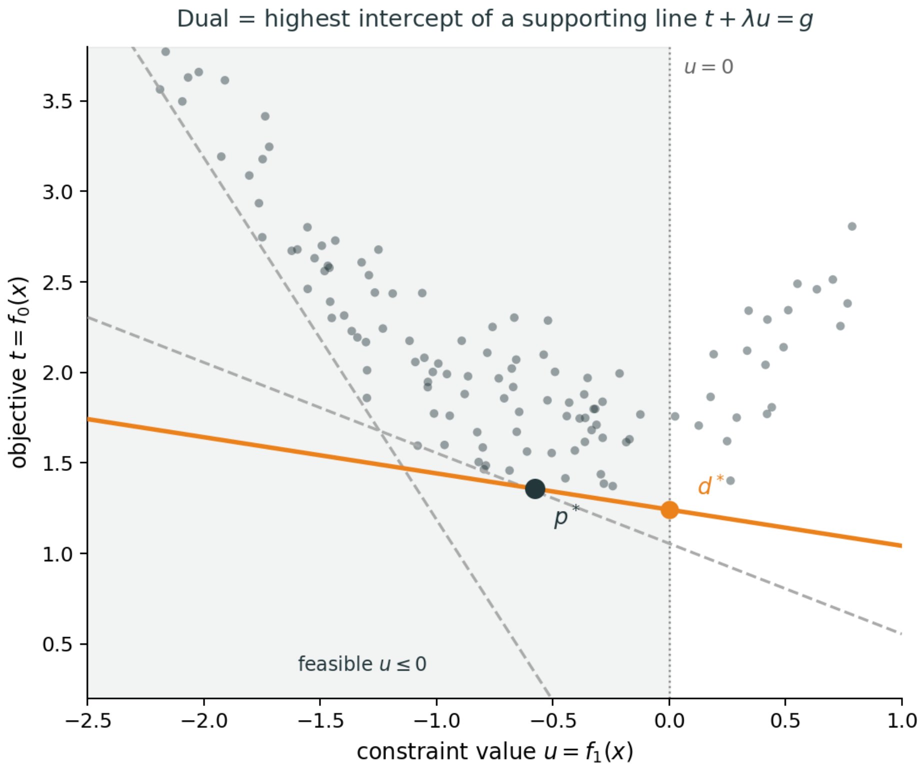
form the **whole cloud** of achievable outcomes, both feasible ($u \le 0$) and infeasible. The primal value p^* is the smallest t among points in $\mathcal{A}$ with $u \le 0$. For $\lambda \ge 0$, lines $t + \lambda u = \alpha$ have slope $-\lambda$ in the (u, t) plane, and

$$g(\lambda) = \inf_{(u,t) \in \mathcal{A}} (t + \lambda u)$$

is the smallest α such that the line still touches $\mathcal{A}$ from below (its intercept on $u = 0$). The infimum is over **all** of $\mathcal{A}$, not only the feasible slice $u \le 0$.

Claim 4 (Geometric weak duality in the (u, t) picture)**.** *For every $\lambda \ge 0$, $g(\lambda) \le p^*$. The dual value $d^* = \sup_{\lambda \ge 0} g(\lambda)$ is the largest intercept at $u = 0$ obtained by a supporting line below $\mathcal{A}$.*

Proof. Take any feasible point $(u, t) \in \mathcal{A}$ with $u \le 0$. Since $\lambda \ge 0$, $t + \lambda u \le t$. Also $g(\lambda) \le t + \lambda u$ by definition of the infimum. Hence $g(\lambda) \le t$. Minimising t over feasible points gives $g(\lambda) \le p^*$. Maximising $g(\lambda)$ over $\lambda \ge 0$ gives d^*. □



Geometry II: one inequality, one equality, and the (u, v, t) cloud. Key idea. Adding an equality lets the supporting object tilt in a **third** direction; free ν can tighten the bound.

Now add $h(\mathbf{x}) = 0$ and write $u = f_1(\mathbf{x})$, $v = h(\mathbf{x})$, $t = f_0(\mathbf{x})$. The cloud

$$\mathcal{A} = \{\, (u, v, t) : \; u = f_1(\mathbf{x}), \; v = h(\mathbf{x}), \; t = f_0(\mathbf{x}), \; \mathbf{x} \in D \,\}$$

again contains **all** achievable triples; feasible points satisfy $u \leq 0$ and $v = 0$. Among them, the primal minimises t, giving p^*. Choose $\lambda \geq 0$ (for u) and ν free (for v). The plane $t + \lambda u + \nu v = \alpha$ supports $\mathcal{A}$ from below, and

$$g(\lambda, \nu) = \inf_{(u,v,t) \in \mathcal{A}} (t + \lambda u + \nu v)$$

is its intercept at $u = 0$, $v = 0$.

Claim 5 (Geometric weak duality in the (u, v, t) picture)**.** *For every $\lambda \geq 0$ and ν, $g(\lambda, \nu) \leq p^*$. The dual problem $d^* = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$ seeks the best intercept from a hyperplane below the full cloud $\mathcal{A}$.*

Proof. For any feasible (u, v, t), $u \leq 0$ and $v = 0$, so $t + \lambda u + \nu v = t + \lambda u \leq t$. Since $g(\lambda, \nu) \leq t + \lambda u + \nu v$, we get $g(\lambda, \nu) \leq p^*$. □

Remark 8 (Why $\nu = 0$ can be weaker)**.** *With $\nu = 0$, the plane $t + \lambda u = \alpha$ cannot tilt in the v-direction. The infimum $g(\lambda, 0)$ may occur at $v \neq 0$, giving a low intercept at $v = 0$. Allowing ν to vary lets the plane tilt so the minimum over $\mathcal{A}$ can occur nearer $v = 0$, often raising the bound; for convex problems this can achieve $d^* = p^*$. For LPs, $\mathcal{A}$ is a polyhedron (convex), so strong duality holds when the primal optimum is finite.*

Having seen the geometry, we verify that the LP dual of Section 2 is exactly what the Lagrangian recipe produces.

LP recovers Section 2.

Claim 6 (LP Lagrangian dual equals the LP dual)**.** *Write* min $\mathbf{c}^\mathsf{T}\mathbf{x}$ *s.t.* $A\mathbf{x} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$ *as* $f_0(\mathbf{x}) = \mathbf{c}^\mathsf{T}\mathbf{x}$, $f_1(\mathbf{x}) = \mathbf{b} - A\mathbf{x} \leq \mathbf{0}$, $f_2(\mathbf{x}) = -\mathbf{x} \leq \mathbf{0}$*. The Lagrangian dual* (2) *is obtained with* $\mathbf{y} = \boldsymbol{\lambda}$*.*

Proof. The Lagrangian is

$$L = \mathbf{c}^\mathsf{T}\mathbf{x} + \boldsymbol{\lambda}^\mathsf{T}(\mathbf{b} - A\mathbf{x}) + \boldsymbol{\mu}^\mathsf{T}(-\mathbf{x}).$$

For fixed $(\boldsymbol{\lambda}, \boldsymbol{\mu})$,

$$g(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} \left[(\mathbf{c} - A^\mathsf{T}\boldsymbol{\lambda} - \boldsymbol{\mu})^\mathsf{T}\mathbf{x} + \boldsymbol{\lambda}^\mathsf{T}\mathbf{b} \right].$$

If $A^\mathsf{T}\boldsymbol{\lambda} + \boldsymbol{\mu} \neq \mathbf{c}$, the infimum is $-\infty$. If $A^\mathsf{T}\boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{c}$, then $g = \boldsymbol{\lambda}^\mathsf{T}\mathbf{b}$. Requiring $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\boldsymbol{\mu} \geq \mathbf{0}$ and writing $A^\mathsf{T}\boldsymbol{\lambda} \leq \mathbf{c}$ gives (2) with $\mathbf{y} = \boldsymbol{\lambda}$. $\square$

When the problem is convex and strictly feasible, weak duality upgrades to strong duality.

Strong duality for convex problems. The problem is **convex** if f_0, f_i are convex, h_j are affine, and D is convex.

Theorem 9 (Strong duality)**.** *If the primal is convex,* ***Slater's condition*** *holds (strict feasibility:* $f_i(\mathbf{x}) < 0$*,* $h_j(\mathbf{x}) = 0$ *for some* $\mathbf{x} \in \operatorname{relint} D$*), and* p^* *is finite, then* $d^* = p^*$ *and the dual optimum is attained.*

Remark 10 (What is relint D?)**.** *The* ***relative interior*** $\operatorname{relint} D$ *is the "interior" of* D *measured inside the flat (affine hull) that contains* D*, not inside all of* $\mathbb{R}^n$*.*

- *If* $D = \mathbb{R}^n$*, then* $\operatorname{relint} D = \mathbb{R}^n$*.*
- *If* D *is a line segment in* $\mathbb{R}^2$*, its usual topological interior is empty, but* $\operatorname{relint} D$ *is the* ***open segment*** *(endpoints excluded).*
- *If* $D = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$ *is a full-dimensional polyhedron,* $\operatorname{relint} D$ *is the set of points that satisfy every inequality* ***strictly*** *(*$A\mathbf{x} < \mathbf{b}$*), i.e. not on a lower-dimensional face.*

Slater asks for a ***strictly feasible*** *point that is not trapped on a boundary of* D*: inequalities hold with slack (*$f_i(\mathbf{x}) < 0$*), equalities hold exactly, and* $\mathbf{x}$ *lies in the relative interior of the domain.*

For LPs, a feasible point with finite optimum suffices. Slater is the convex analogue of "interior point exists."

Economics. λ_i, ν_j are shadow prices on constraints. The dual finds the price system giving the best accounting lower bound on optimal cost. When the gap vanishes, prices fully explain the optimum.